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Flexible display panel, display device and forming method

Abstract

A flexible display panel, a display device and a forming method are provided. The flexible display panel includes a display area and a non-display area, where the non-display area includes a bending sub-area and a binding sub-area, and the bending sub-area is configured to bend the binding sub-area to a side away from the display area, the bending sub-area includes a first organic layer, a second organic layer and a metal wiring layer between the first organic layer and the second organic layer arranged on the substrate, the touch wiring layer is electrically connected to the metal wiring layer; in response to the bending sub-area being in a bending state, a vertical distance from a bending neutral layer of the bending sub-area to the metal wiring layer is smaller than a preset distance.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11974475	12/2023	Wei	N/A	G09F 9/33
2014/0240985	12/2013	Kim et al.	N/A	N/A
2017/0262109	12/2016	Choi	N/A	G06F 3/04164
2018/0151662	12/2017	Rhe	N/A	H10K 59/40
2018/0331319	12/2017	Hiraga et al.	N/A	N/A
2019/0095007	12/2018	Jeong et al.	N/A	N/A
2019/0096974	12/2018	Kim	N/A	N/A
2021/0184139	12/2020	Zhang et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
109560109	12/2018	CN	N/A
109585494	12/2018	CN	N/A
109755256	12/2018	CN	N/A
109817675	12/2018	CN	N/A
110197845	12/2018	CN	N/A
110515499	12/2018	CN	N/A

OTHER PUBLICATIONS

Non-final Office Action for copending U.S. Appl. No. 17/425,222, dated Jul. 20, 2023, 6 Pages. cited by applicant

International Search Report and Written Opinion for Application No. PCT/CN2020/077264, dated Nov. 27, 2020, 9 Pages. cited by applicant

Notice of Allowance for copending U.S. Appl. No. 17/425,222, dated Jan. 4, 2024, 9 Pages. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/425,222 filed on Jul. 22, 2021 which is the U.S. national phase of PCT Application PCT/CN2020/077264 filed on Feb. 28, 2020, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD
(1) The disclosure relates to the technical field of flexible display, and in particular to a flexible display panel, a display device and a forming method.
BACKGROUND
(2) Organic Light-Emitting Diode (OLED) is a new display technology, and has the characteristics of active light emission, high brightness, wide viewing angle, fast response speed, low energy consumption, flexibility and the like, so that the Organic Light-Emitting Diode (OLED) receives more and more attention, and becomes a next generation display technology capable of replacing liquid crystal display.
(3) The flexible OLED display technology is becoming mature continuously, and the types of flexible display products are diversified continuously; frameless product has become a market mainstream. In order to save drive division's occupation space, improve the screen-to-bod ratio, reach comprehensive screen display effect, when assembling the flexible OLED display panel in the related art, a part of the driving area of the display panel is bent to the back of the display panel. However, in the process of bending the bending area, the metal wires in the bending area are easily broken due to the stress, thereby causing wire defects.
SUMMARY
(4) The purpose of the embodiments of the present disclosure is to provide a flexible display panel, a display device, and a forming method, which solve the problem that when a driving portion of the flexible display panel in the prior art is bent to the back of the display panel, a metal trace is easily broken.
(5) In order to achieve the above object, the present disclosure provides a flexible display panel, which includes a display area and a non-display area, where the non-display area includes a bending sub-area and a binding sub-area, and the bending sub-area is configured to bend the binding sub-area to a side away from the display area; the display area includes a driving circuit layer arranged on a substrate, the driving circuit layer includes a source/drain electrode layer, a planarization layer arranged at a side, far away from the substrate, of the source/drain electrode layer, a pixel defining layer arranged at a side, far away from the substrate, of the planarization layer and a touch wiring layer arranged at a side, far away from the substrate, of the pixel defining layer, the bending sub-area includes a first organic layer, a second organic layer and a metal wiring layer between the first organic layer and the second organic layer arranged on the substrate, the touch wiring layer is electrically connected to the metal wiring layer, and an orthographic projection of the touch wiring layer on a plane of the substrate is not located in the bending sub-area; in response to the bending sub-area being in a bending state, a vertical distance from a bending neutral layer of the bending sub-area to the metal wiring layer is smaller than a preset distance.
(6) Optionally, the metal wiring layer is arranged in a same layer as the source/drain electrode layer.
(7) Optionally, the display region further includes a first electrode within the pixel defining layer, a light-emitting layer at a side of the first electrode away from the substrate and a second electrode at

a side of the light-emitting layer away from the substrate; the first electrode is connected to the source/drain electrode layer through a transfer metal layer, and the metal wiring layer and the transfer metal layer are arranged on the same layer.

(8) Optionally, the planarization layer includes a first planarization layer and a second planarization layer, where the transfer metal layer is on the first planarization layer and within the second planarization layer, and the transfer metal layer is connected to the source/drain electrode layer through a first via hole penetrating through the first planarization layer; the first electrode is connected to the transfer metal layer through a second via hole penetrating through the second planarization layer.

(9) Optionally, the first organic layer is closer to the substrate than the second organic layer, the first organic layer is in the same layer as the first planarization layer, the second organic layer is in the same layer as the second planarization layer and the pixel defining layer.

(10) Optionally, the bending neutral layer overlaps the metal wiring layer in response to the bending sub-area being in a bent state.

(11) Optionally, the metal wiring layer is located at one-half of a thickness of the bending sub-area in response to the bending sub-area being not in the bent state.

(12) Optionally, a thickness adjusting layer is at a side of the second organic layer away from the metal wiring layer in the bending sub-area; through the thickness adjusting layer, in response to the bending sub-area being in a bending state, a vertical distance from the bending neutral layer of the bending sub-area to the metal wiring layer is smaller than a preset distance.

(13) Optionally, the thickness adjusting layer further extends to the display area and is located at a side of the touch wiring layer away from the substrate.

(14) Optionally, the thickness adjusting layer is an organic material layer.

(15) Optionally, the substrate includes a first organic material, a second organic material, and at least one inorganic layer between the first organic material and the second organic material.

(16) Optionally, the bending neutral layer is located at a side of the metal wiring layer proximate to the substrate.

(17) Optionally, the preset distance is less than or equal to 5 microns.

(18) A display device including the flexible display panel above is further provided in the present disclosure.

(19) Optionally, a driving component is arranged on the binding sub-area on the non-display region, and the metal wiring layer is electrically connected to the driving component; the bending sub-area is bent relative to the display area, and the driving component is arranged at a side away from the display area.

(20) A method of forming a flexible display panel above is further provided in the present disclosure, including: providing a substrate; and forming a display area and a non-display area on the substrate; where the non-display area includes a bending sub-area and a binding sub-area, and the bending sub-area is configured to bend the binding sub-area to a side away from the display area; the display area includes a driving circuit layer arranged on a substrate, the driving circuit layer includes a source/drain electrode layer, a planarization layer arranged at a side, far away from the substrate, of the source/drain electrode layer, a pixel defining layer arranged at a side, far away from the substrate, of the planarization layer and a touch wiring layer arranged at a side, far away from the substrate, of the pixel defining layer, the bending sub-area includes a first organic layer, a second organic layer and a metal wiring layer between the first organic layer and the second organic layer arranged on the substrate, the touch wiring layer is electrically connected to the metal wiring layer, and an orthographic projection of the touch wiring layer on a plane of the substrate is not located in the bending sub-area; in response to the bending sub-area being in a bending state, a vertical distance from a bending neutral layer of the bending sub-area to the metal wiring layer is smaller than a preset distance.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to more clearly illustrate the technical solutions in the present embodiment or related technologies of the present disclosure, the drawings used in the description of the embodiments will be briefly introduced below, it is obvious that the drawings in the following description are only some embodiments of the present disclosure, and it is obvious for those skilled in the art that other drawings may be obtained according to the drawings without creative efforts.

(2) FIG. **1** is a schematic view of an unfolded state of a flexible display substrate according to the present disclosure;

(3) FIG. **2** is a schematic view of a curved state of a flexible display substrate according to the present disclosure;

(4) FIG. **3** is a schematic cross-sectional view of a bending sub-area in a bending state according to an embodiment of the disclosure;

(5) FIG. **4** is a second schematic cross-sectional view of a bending sub-area in a bending state according to an embodiment of the disclosure;

(6) FIG. **5** is a third schematic cross-sectional view of a bending sub-area in a bending state according to an embodiment of the disclosure;

(7) FIG. **6** is a schematic cross-sectional view of a bending sub-area in a planar state according to an embodiment of the present disclosure;

(8) FIG. **7** is a third schematic cross-sectional view of a bending sub-area in a bending state according to an embodiment of the disclosure;

(9) FIG. **8** is a fourth schematic cross-sectional view of a bending sub-area in a bending state according to an embodiment of the present disclosure;

(10) FIG. **9** is a schematic cross-sectional view of a flexible display panel according to the present disclosure;

(11) FIG. **10** is a second schematic cross-sectional view of a flexible display panel according to the present disclosure;

(12) FIG. **11** is a schematic plan view of a flexible display panel according to another embodiment of the present disclosure;

(13) FIG. **12** is a schematic sectional view taken along line I-I in FIG. **11**;

(14) FIG. **13** is a cross-sectional view of part B-B of FIG. **11**;

(15) FIG. **14** is a second cross-sectional view of the portion B-B in FIG. **11**;

(16) FIG. **15** is a second cross-sectional view of the portion B-B in FIG. **11**.

DETAILED DESCRIPTION

(17) The technical solutions in the embodiments of the present disclosure will be described below clearly and completely with reference to the accompanying drawings in the embodiments of the present disclosure, and it is obvious that the described embodiments are only some embodiments of the present disclosure, not all embodiments. All other embodiments, which may be derived by a person skilled in the art from the embodiments disclosed herein without making any creative effort, shall fall within the scope of the present disclosure.

(18) The main purpose of the embodiments of the present disclosure is to provide a flexible display panel, a display device, and a forming method, which solve the problem that when a driving portion of the flexible display panel in the prior art is bent to the back of the display panel, a metal trace is easily broken.

(19) The flexible display panel according to the embodiment of the present disclosure, as shown in FIGS. **1** and **2**, includes a display area **100** and a non-display area **200** at one end of the display area **100**. In the direction from the display area **100** to the display area **100**, the non-display area **200** includes a first transition sub-area **201**, a bending sub-area **202**, a second transition sub-area

203, and a binding sub-area **204**, where the bending sub-area **202** includes a planar state (as shown in FIG. **1**) located on the same plane as the display area **100**, and a bending state (as shown in FIG. **2**) bent relative to the display area **100**.
(20) The flexible display panel comprises a substrate **1**, where a display function device is formed on the substrate **1** and used for realizing a display function, and the display function device is correspondingly formed into a display area **100** in an area arranged on the substrate **1**; on the substrate **1**, the area where the display function device is not arranged is a non-display area **200**, where the driving component **300** is arranged on the binding sub-area **204** of the non-display area **200**. The bending sub-area **202** is located between the binding sub-area **204** and the display region **100**, and is configured to bend the binding sub-area **204** to a side away from the display region **100**, so that the driving component **300** on the binding sub-area **204** is connected to the flexible circuit board of the display device.
(21) In the embodiment of the present disclosure, the substrate **1** is a flexible transparent substrate, and can realize bending of the display panel.
(22) In the embodiment of the present disclosure, the driving component **300** includes, but is not limited to, a driving chip and a flexible circuit board, and a metal wiring layer is arranged on the bending sub-area **202** and electrically connected to the driving component **300**, and the metal wiring layer can transmit a control signal on the driving component **300** to the display area **100** to control image display or touch operation of the display area **100**.
(23) Referring to FIG. **2**, in the flexible display panel according to the embodiment of the present disclosure, when the bending sub-area **202** is bent toward a side away from the display region **100** with respect to the display region **100**, and is in a bending state, the driving component **300** located on the non-display region **200** and far away from the display region **100** may be bent to the back of the display region **100**, so as to complete the next module assembly, thereby reducing the display frame of the display region **100** of the flexible display panel and achieving the effect of narrow-frame display. In order to solve the issue that, in the process of assembling the flexible display panel, when bending the sub-area **202**, the sub-area **202** may be subjected to a great stress, and the stress concentrates on the metal wiring layer of sub-area **202** that being bent, cause the metal wiring layer to be broken easily, this embodiment of the disclosure flexible display panel, as shown in FIG. **3**, when the bending sub-area **202** is in the state of bending, by making a vertical distance h between the metal wiring layer **211** and the bending neutral layer **212** to be less than preset distance, the metal wiring layer **211** may be close to the bending neutral layer **212** as far as possible, thereby reducing the stress on the metal wiring layer **211** of the bending sub-area **202**, avoiding metal wiring layer **211** from being broken.
(24) Optionally, the preset distance is less than or equal to 5 microns.
(25) It should be noted that the principle of setting the preset distance may be as follows: the bending test experiment is performed on the bending sub-area **202**, and the critical value of the preset distance that can satisfy the preset test condition is the set value of the preset distance. It is understood that, depending on the total thickness of the bending sub-area **202**, the material and thickness of each film layer on the bending sub-area **202**, there is a corresponding difference in the preset distance.
(26) In one embodiment of the present disclosure, as shown in FIG. **3**, a vertical distance h from the metal wiring layer **211** to the bending neutral layer **212** may be defined as a vertical distance from a central plane **2111** of the metal wiring layer **211** to the bending neutral layer **212**.
(27) In the disclosed embodiment, as shown in FIG. **3**, a central plane **2111** of metal wiring layer **211** is located between opposing first and second side surfaces **2112**, **2113** of metal wiring layer **211**, and is equidistant from first and second side surfaces **2112**, **2113**.
(28) In particular, it may be appreciated that the metal wiring layer **211** is formed as a thin layer structure, the first side surface **2112** and the second side surface **2113** are two opposite sides of the metal wiring layer **211** forming a side of the layer structure, and in any bending state, the central

plane **2111** is a plane parallel to the first side surface **2112** and the second side surface **2113** between the first side surface **2112** and the second side surface **2113**, and is equidistant from the first side surface **2112** and the second side surface **2113**.
(29) In the embodiment of the present disclosure, the vertical distance from the central plane **2111** to the bending neutral layer **212** includes the vertical distance from each position point on the central plane **2111** to the bending neutral layer **212**, and in the flexible display panel, the vertical distance h from each position point on the central plane **2111** to the bending neutral layer **212** is less than the preset distance.
(30) In another embodiment of the present disclosure, as shown in FIG. **4**, since the metal wiring layer **211** is generally formed as a thin layered structure, without precise requirement, the vertical distance h from the metal wiring layer **211** to the bending neutral layer **212** may also be defined as the vertical distance from the side surface of the metal wiring layer **211** close to the bending neutral layer **212**. For example, when the bending neutral layer **212** is closer to the second side surface **2113** of the metal wiring layer **211** than to the first side surface **2112**, the vertical distance h from the metal wiring layer **211** to the bending neutral layer **212** may be defined as the distance from the second side surface **2113** of the metal wiring layer **211** to the bending neutral layer **212**. It should be noted that, as shown in FIG. **2**, FIG. **3** and FIG. **4**, when bending sub-area **202** is in a bending state, a compressive stress is generated on the first surface **2101**, and a tensile stress is generated on the opposite second surface **2102**, based on which at each position point in the direction from the first surface **2101** to the second surface **2102** on the cross section of bending sub-area **202**, the compressive stress is gradually transited to the tensile stress, and therefore, there are inevitably position points which are not subjected to the compressive stress or the tensile stress, and the planes of the position points are bending neutral layers **212**. When the bending sub-area **202** is in a bent state, the bending neutral layer **212** will not be deformed.
(31) Further, the bending sub-areas **202** have different deformation degrees when in different bending states, where the positions of the bending neutral layers **212** in different deformation degrees are slightly different, in the flexible display panel according to the embodiment of the present disclosure, when the bending sub-areas **202** are in any bending state that is bent away from a side of the display region **100**, the vertical distance h from the metal wiring layer **211** to the bending neutral layers **212** is smaller than the preset distance, so as to effectively solve the problem that the metal wiring is easily broken when the bending sub-areas **202** of the flexible display panel are bent to the back of the display panel. In one embodiment of the present disclosure, as shown in FIG. **5**, when the bending sub-area **202** is in the bending state, the bending neutral layer **212** overlaps the metal wiring layer **211**. That is, the bending neutral layer **212** is located on the metal wiring layer **211**, and the distance from the metal wiring layer **211** to the bending neutral layer **212** is zero.
(32) Specifically, by locating the bending neutral layer **212** on the metal wiring layer **211**, a tensile stress or a compressive stress on the metal wiring layer **211** caused by bending the bending sub-area **202** may be avoided, so that the metal wiring layer **211** may be effectively prevented from being broken.
(33) In one embodiment of the present disclosure, as shown in FIG. **1** and FIG. **6**, the bending sub-area **202** further includes a planar state of the same layer as the display area **100**, and in the planar state, the central plane **2111** of the metal wiring layer **211** is located at one-half the thickness of the bending sub-area **202**.
(34) By having the center plane **2111** of the metal wiring layer **211** located at one-half the thickness of the bending sub-area **202** in the planar state, as shown in FIG. **7**, it is ensured that the center plane **2111** of the metal wiring layer **211** may be closer to the bending neutral layer **212** when the bending sub-area **202** is bent.
(35) In one embodiment of the flexible display panel of the present disclosure, the relative position of the metal wiring layer **211** between the first surface **2101** and the second surface **2102** of the

bending sub-area **202** may be adjusted by setting the total thickness of the bending sub-area **202**, so as to satisfy at least one of the following implementation structures: the vertical distance from the metal wiring layer **211** to the bent neutral layer **212** is less than a preset distance; the neutral layer **212** is overlapped with the metal wiring layer **211**; in the planar state, the central plane **2111** of the metal wiring layer **211** is located at one-half of the thickness of the bending sub-area **202**.
(36) In one embodiment of the flexible display panel of the present disclosure, as shown in FIG. **8**, a substrate **1** is arranged at a side of a metal wiring layer **211** in a bending sub-area **202**, a first organic layer **401**, the metal wiring layer **211**, and a second organic layer **402** are sequentially arranged on the substrate **1**, and a thickness adjusting layer **500** is further arranged at a side of the second organic layer **402** away from the substrate **1**; through the thickness adjusting layer **500**, in the bending state, the vertical distance from the metal wiring layer **211** to the bending neutral layer **212** is less than a preset distance.
(37) Specifically, on the basis that the first organic layer **401**, the metal wiring layer **211** and the second organic layer **402** are arranged in the bending sub-area **202**, the thickness adjusting layer **500** is arranged to adjust the total thickness of the bending sub-area **202** and the relative position of the metal wiring layer **211** between the first surface **2101** and the second surface **2102** of the bending sub-area **202**, so as to ensure that the vertical distance from the metal wiring layer **211** to the bending neutral layer **212** is less than the preset distance.
(38) The following describes a specific implementation structure of the flexible display panel according to an embodiment of the present invention in detail with reference to specific embodiments.
(39) As shown in FIG. **9**, in the flexible display panel according to one embodiment of the present disclosure, taking the flexible display panel as a top emission OLED display panel and having a touch function as an example, the flexible display panel includes a display area **100** and a non-display area **200**, where the non-display area **200** includes a bending sub-area **202** and a binding sub-area, where the binding sub-area is not shown in FIG. **9**, and referring to FIG. **1**, the binding sub-area is located at a side of the bending sub-area **202** away from the display area **100**, and the bending sub-area **202** is used to bend the binding sub-area to a side away from the display area **100**; the display area **100** comprises a driving circuit layer arranged on a substrate **1**, the driving circuit layer comprises a source/drain electrode layer **605**, a planarization layer **607** arranged at a side, far away from the substrate **1**, of the source/drain electrode layer **605**, a pixel defining layer **703** arranged at a side, far away from the substrate **1**, of the planarization layer **607**, and a touch wiring layer **900** arranged at a side, far away from the substrate **1**, of the pixel defining layer **703**, the bending sub-area **202** comprises a first organic layer **401** arranged on the substrate **1**, a second organic layer **402**, and a metal wiring layer **211** arranged between the first organic layer **401** and the second organic layer **402**, the touch wiring layer **900** is electrically connected to the metal wiring layer **211**, and the orthographic projection of the touch wiring layer **900** on the plane of the substrate **1** is not located in the bending sub-area **202**; when the bending sub-area **202** is in a bending state, a vertical distance from a bending neutral layer of the bending sub-area **202** to the metal wiring layer is smaller than a preset distance.
(40) Further, specifically, the flexible display panel further includes an active layer **601**, a gate insulating layer **602**, a gate **603**, and an interlayer insulating layer **604** sequentially formed on the substrate **1**, where a source/drain layer **605** is formed on the interlayer insulating layer **604**; and further comprising a light emitting device layer between the driver circuit layer and the touch wiring layer **900**, the light emitting device layer comprising a first electrode **702**, a light emitting layer **704** arranged at a side of the first electrode **702** remote from the substrate **1**, and a second electrode **705** arranged at a side of the light emitting layer **704** remote from the substrate **1**; optionally, a spacer layer **706** is further arranged on the second electrode **705**. Here, a pixel defining layer **703** is arranged on the first electrode **702**, and a light emitting layer **704** is arranged in the pixel defining layer **703**. Optionally, the spacer layer **706** includes two inorganic layers

opposite to each other and an organic layer between the two inorganic layers.
(41) The first electrode **702** is connected to the source/drain layer **605** through a via hole of the planarization layer **607**, and the light emitting device may be driven to emit light through the driving circuit layer.
(42) In the embodiment of the present disclosure, optionally, the first electrode **702** is an anode, and the second electrode **705** is a cathode. The light emitting layer **704** includes a hole injection layer, a hole transport layer, a light emitting layer, an electron transport layer, and an electron injection layer, which are sequentially arranged. In the embodiment of the present disclosure, optionally, the touch wiring layer **900** is arranged on the spacer layer **706**, and optionally, the flexible display panel further includes an encapsulation layer **800** arranged on the touch wiring layer **900**.
(43) In this embodiment, the touch trace layer **900** on the display area **100** is connected to the metal trace layer **211** on the bending sub-area **202**, so that the touch signal lines may be transmitted to the driving elements on the binding sub-area through the metal trace layer **211**.
(44) In the embodiment of the present disclosure, in order to ensure that the perpendicular distance from the bending neutral layer of the bending sub-area **202** to the metal wiring layer **211** is smaller than the preset distance when the bending sub-area **202** is in the bent state, optionally, the total thickness of the bending sub-area **202** may be adjusted by adjusting the thickness of the first organic layer **401** and/or the second organic layer **402**, so as to adjust the thickness from the metal wiring layer **211** to the bending neutral layer of the bending sub-area **202**.
(45) In particular, the positional relationship between the metal wiring layer **211** and the bending neutral layer of the bending sub-area **202** may be found in conjunction with FIGS. **1** to **8** and with reference to the above detailed description.
(46) In another embodiment of the present disclosure, referring to FIG. **10**, the display area **100** of the flexible display panel includes the driving circuit layer and the light emitting device layer of the above-mentioned implementation structure, and the bending sub-area **202** includes the first organic layer **401**, the second organic layer **402** and the metal wiring layer **211** located between the first organic layer **401** and the second organic layer **402**, which are arranged on the substrate **1**, on the basis of the arrangement structure of the touch wiring layer **900** electrically connected to the metal wiring layer **211**, the bending sub-area **202**, the metal wiring layer **211** and the source/drain electrode layer **605** are arranged in the same layer, the first organic layer **401** located between the metal wiring layer **211** and the substrate **1** may be arranged in the same layer with the planarization layer **607**, and the second organic layer **402** may be arranged in the same layer with the pixel defining layer **703**.
(47) Based on the above embodiment, since the thicknesses of the interlayer insulating layer **604**, the gate insulating layer **602**, the pixel defining layer **703** and the planarization layer **607** are generally within a predetermined range in the fabrication of the display device, for example, the substrate **1** is about 100000 angstroms, the gate insulating layer **602** is about 1200 to 1400 angstroms, the interlayer insulating layer **604** is about 4000 to 6000 angstroms, the pixel defining layer **703** is generally 14000 to 16000 angstroms, and the planarization layer **607** is generally 20000 angstroms, since the thickness of the substrate **1** is much greater than the thickness of each layer arranged on the substrate, in order to ensure that the perpendicular distance from the bending neutral layer of the bending sub-area **202** to the metal wiring layer **211** in the bending sub-area **202** is smaller than a preset distance, optionally, a thickness adjusting layer **500** is further arranged on the second organic layer **402** away from the substrate **1** on the bending sub-area **202**, and the overall thickness of the bending sub-area **202** is adjusted by the thickness adjusting layer **500**.
(48) In one embodiment of the present disclosure, optionally, the encapsulation layer **800** of the display area **100** extends to the bending sub-area **202**, is arranged on the second organic layer **402** of the bending sub-area **202**, and is formed as the thickness adjusting layer **500** of the bending sub-area **202**.
(49) Optionally, the encapsulation layer **800** includes a first inorganic layer, an organic layer, and a

second inorganic layer stacked in this order. In this embodiment, when the first organic layer **401** and the second organic layer **402** of the bending sub-area **202** respectively have the same layer as the corresponding layer of the display region **100**, since the total thickness of the substrate **1** is usually 3 times of the total thickness of the first organic layer **401**, the metal wiring layer **211** and the second organic layer **402** arranged above, the bending neutral layer **212** (shown in FIGS. **1** to **8**) of the bending sub-area **202** is usually located on the substrate **1**, or located at a longer distance at the side of the metal wiring layer **211** close to the substrate **1**. On the basis of ensuring the realization of the process, by the thickness adjusting layer **500**, the bending neutral layer **212** of the bending sub-area **202** is still located at the side of the metal wiring layer **211** facing the substrate **1**.
(50) The flexible display substrate according to the above-described embodiment of the present disclosure may be formed such that the layers of the non-display region **200** are simultaneously formed when the driving circuit layer and the light emitting device layer are formed in the display region of the substrate **1**.
(51) In addition, in the embodiment of the present disclosure, the thickness adjusting layer **500** of the bending sub-area **202** is not limited to be formed by extending the encapsulation layer **800** through the display region **100**. For example, the total thickness of the bending sub-area **202** may be adjusted by separately preparing the thickness adjusting layer **500** in the bending sub-area **202**, so that the metal wiring layer **211** is close to the bending neutral layer **212** of the bending sub-area **202**.
(52) Optionally, a separately prepared thickness adjusting layer **500** is located at a side of the second organic layer **402** away from the substrate **1**, and after the encapsulation layer **800** is prepared in the display area **100**, a step of further preparing the thickness adjusting layer **500** on the second organic layer **402** of the non-display area **200** farthest from the substrate **1** is further included based on the arrangement.
(53) Optionally, the thickness adjusting layer **500** of the bending sub-area **202** may also extend to the display region **100** and be located at a side of the touch wiring layer **900** away from the substrate **1**.
(54) In one embodiment, the thickness adjusting layer **500** is an organic material layer.
(55) In the above embodiments of the present disclosure, the thickness adjusting layer **500** is arranged at a side of the second organic layer **402** away from the metal wiring layer **211** in the bending sub-area **202**. The thickness adjusting layer **500** is not limited to be only at this position, and may be between the first organic layer **401** and the metal wiring layer **211**, or between the second organic layer **402** and the metal wiring layer **211**, for example.
(56) According to another embodiment of the flexible display substrate disclosed by the disclosure, according to the specific structures of the display area and the non-display area, a structure in which the vertical distance from the central plane of the metal wiring layer to the bending neutral layer is less than the preset distance is realized, and the structure is not limited to a mode of only increasing the thickness adjusting layer, and for example, the structure can also be realized by reasonably setting the thickness of the extension layer of each display area on the non-display area.
(57) In the above embodiments of the present disclosure, a specific implementation structure of the flexible display substrate of the present disclosure is described by taking an example that the source/drain layer of the display area extends to the non-display area, the source/drain layer extending to the non-display area is formed as a metal wiring layer of the non-display area, and a vertical distance from a central plane of the metal wiring layer of the non-display area to the bending neutral layer is smaller than a preset distance by providing a thickness adjusting layer.
(58) Further, the flexible display panel of the present disclosure is not limited to being applicable only to the above-described implementation structure.
(59) For example, in another embodiment of the flexible display panel according to the present disclosure, the flexible display panel is a display panel with a touch function, and a touch wiring layer is arranged in the display area, as shown in FIG. **11**, in the display area **100**, the touch wiring

layer includes a plurality of first touch electrodes **921** arranged along a first direction and a plurality of second touch electrodes **922** arranged along a second direction.
(60) According to FIG. **11**, in the embodiment of the present disclosure, at one edge of the display area **100**, the flexible display panel extends to form a non-display area **200** in a direction away from the edge, and from a direction close to the display area **100** to a direction away from the display area **100**, the non-display area **200** includes a first transition sub-area **201**, a bending sub-area **202**, a second transition sub-area **203**, and a binding sub-area **204**, where the binding sub-area **204** is with a driving component **300**.
(61) Optionally, the drive assembly **300** includes a control chip **310** and a plurality of line terminals **320**. Some of the plurality of line terminals **320** are connected to the first touch electrodes **921** of the display area **100** arranged along the first direction through connection lines, for inputting touch scan signals to the first touch electrodes **921**; another portion of the plurality of line terminals **320** is connected to the second touch electrodes **922** arranged along the second direction of the display area **100** through the connection lines, for obtaining touch sensing signals on the second touch electrodes **922**. The first touch electrodes **921** arranged along the first direction and the second touch electrodes **922** arranged along the second direction are intersected with each other, and a touch scanning signal is input to the plurality of first touch electrodes **921** according to a preset frequency to obtain an induction signal on each second touch electrode **922**, so that a touch operation position on the flexible display panel may be determined.
(62) In the embodiment of the present disclosure, the region of the non-display area **200** where the driving element **300** is arranged is the binding sub-area **204**, and on the non-display area **200**, the bending sub-area **202** is located between the binding sub-area **204** and the display area **100**. Through the bending sub-area **202**, the binding sub-area **204** may be bent to a state opposite to the display region **100**, so that the driving element **300** arranged on the binding sub-area **204** is opposite to the display region **100**.
(63) In the embodiment of the present disclosure, as shown in FIG. **11**, a first blocking structure **410** and a second blocking structure **420** are further arranged around the display area **100** in the non-display area **200**, the first blocking structure **410** is separated from the second blocking structure **420**, and are formed in a ring shape arranged around the display area **100** at the periphery of the display area **100**, so as to prevent the organic material in the display area **100** from overflowing when the flexible display panel is formed.
(64) Further, on the non-display area **200**, the portions of the first blocking structure **410** and the second blocking structure **420** surrounding the edge of the display area **100** close to the bending sub-area **202** are located on the first transition sub-area **201**.
(65) In one embodiment, a metal wiring layer is arranged on the bending sub-area **202**, and the metal wiring layer may be connected to a power line of the display region **100**, a data line of the display region **100**, the first touch electrode **921** of the display region **100**, or the second touch electrode **922** of the display region, so that the power line, the data line, and the touch wiring are connected to the driving component **300** through the metal wiring layer.
(66) Since the thickness of each layer in the display region is typically in a predetermined range during the fabrication of the display device, the thickness of the inorganic layer is in the range of 1000 to 6000 angstroms, the thickness of the organic layer is in the range of 10000 to 25000 angstroms, for example, in a display device, the substrate **1** is about 90000 angstroms to about 110000 angstroms, e.g., 100000 angstroms, the gate insulating layer **602** is about 1200 angstroms to about 1400 angstroms, such as 1300 angstroms, the interlayer dielectric layer **604** is about 4900 to 5100 angstroms, e.g., 5000 angstroms, the pixel definition layer **703** is typically 14000 to 16000, e.g. 15000 a, the planarization layer **607** is typically 19000-21000 a, e.g. 20000 a, in the bending sub-area **202**, since the thickness of the substrate **1** is much larger than the thickness of the layers on the substrate, in order to ensure that, in the meander sub-area **202**, the perpendicular distance of the meander neutral layer of the meander sub-area **202** to the metal wiring layer **211** is smaller than

a preset distance, optionally, on the bending sub-area **202**, the overall thickness of the bending sub-area **202** is adjusted by a thickness adjusting layer.
(67) Therefore, the thickness adjusting layer is arranged on the bending sub-area **202** of the flexible display panel, and the vertical distance between the metal wiring layer and the bending neutral layer of the bending sub-area **202** is smaller than the preset distance by arranging the thickness adjusting layer, so as to avoid the problem that the metal wiring layer of the bending sub-area **202** is easy to break.
(68) Referring to FIGS. **12** and **13** in combination with FIG. **11**, a display area **100** of the flexible display panel includes a substrate **1**, and a barrier layer **2** and a buffer layer **3** sequentially stacked on the substrate **1** from bottom to top.
(69) The buffer layer **3** is with a driving circuit layer, the driving circuit layer includes a plurality of thin film transistor TFTs, and the TFTs include a first gate insulating layer **6021**, an active layer **601**, a second gate insulating layer **6022**, a gate **603**, an interlayer insulating layer **604**, and a source/drain layer **605**, which are sequentially stacked on the buffer layer **3** from bottom to top. Further, in this embodiment, the active layer **601** is polarized as an electric conductor, and is connected to the source and drain electrodes of the thin film transistor TFT, respectively, to enhance the sensitivity of the control of the thin film transistor TFT.
(70) Optionally, in the embodiment of the present disclosure, the driving circuit layer may further include an inorganic insulating layer, such as silicon nitride or silicon oxide, covering the source/drain layer **605** for protecting the source/drain layer **605** from water and oxygen.
(71) In this embodiment, optionally, a first electrode plate **122** and a second electrode plate **124** are further in the second gate insulating layer **6022** and the interlayer insulating layer **604** to form a storage capacitor, respectively.
(72) Further, the flexible display panel of the present disclosure further includes a light emitting device layer on the driving layer, where a transition metal layer **140** is arranged between the light emitting device layer and the driving layer. Specifically, as shown in FIG. **12**, a first planarization layer **141**, a transition metal layer **140**, and a second planarization layer **142** are sequentially formed on the thin film transistor TFT from bottom to top; the first planarization layer **141** is with a via hole, and the transfer metal layer **140** penetrates through the via hole to be electrically connected to the source or the drain of the thin film transistor TFT.
(73) Optionally, the transfer metal layer **140** may be made of the same material as the source/drain layer **605**, and the flexible display panel may also include an inorganic passivation layer PVX, such as silicon nitride or silicon oxide, covering the transfer metal layer **140** for protecting the transfer metal layer **140** from water and oxygen.
(74) In addition, the light emitting device layer includes: a first pixel defining layer **7031** and a second pixel defining layer **7032** which are sequentially arranged on the second planarization layer **142**; a first electrode **702** and a light-emitting layer **704** in the first pixel defining layer **7031** and the second pixel defining layer **7032**; a second electrode **705** arranged on the second pixel defining layer **7032**.
(75) Further, the flexible display panel is with a spacer layer **706** on the light emitting device layer, and optionally, the spacer layer **706** includes at least three sub-layers.
(76) Referring to FIG. **12**, the flexible display panel further includes a touch wiring layer **900** arranged on the spacer layer **706**, the touch wiring layer **900** includes an insulating layer **910** and a first organic layer **920** arranged on the insulating layer **910**, where a plurality of first touch electrodes **921** and second touch electrodes **922** are formed in the first organic layer **920**.
(77) In the flexible display panel with the above-described implementation structure, the touch wiring layer **900** is further formed above the light emitting device layer, so as to implement the touch function of the flexible display panel.
(78) It should be noted that the first touch electrode **921** and the second touch electrode **922** on the touch wiring layer **900** may be transparent electrode blocks, or may be in a metal grid structure. In

addition, when the first touch electrode **921** and the second touch electrode **922** are respectively an electrode block, one electrode block of the first touch electrode **921** and the second touch electrode **922** corresponds to one light emitting layer **704** of the light emitting device layer, or one electrode block corresponds to a plurality of light emitting layers **704**, and the present disclosure is not limited thereto, which may be adjusted by a person skilled in the art according to the touch precision. FIG. **11** only schematically shows the correspondence between the first touch electrode **921** and the second touch electrode **922**, and the light-emitting layer **704**. With reference to FIG. **11** and FIG. **13**, based on the structure of the display area **100** of the flexible display panel in the foregoing embodiment, in the flexible display panel of this embodiment, the non-display area **200** includes a first transition sub-area **201**, a bending sub-area **202**, a second transition sub-area **203**, and a binding sub-area **204**, which are sequentially arranged from the display area **100** to the edge direction of the flexible display panel.
(79) Referring to FIG. **13**, in the embodiment of the present disclosure, the first transition sub-area **201** includes the substrate **1**, the barrier layer **2**, and the buffer layer **3** in the same layers as the substrate **1**, the barrier layer **2**, and the buffer layer **3** of the display region **100**, respectively. Optionally, the barrier layer **2** and the buffer layer **3** are inorganic layers, and are made of silicon nitride or silicon oxide materials.
(80) Further, the first transition sub-area **201** further includes at least two insulating layers located on the buffer layer **3**, the at least two insulating layers are respectively in the same layer as the first gate insulating layer **6021** and the second gate insulating layer **6022** of the display region **100**, a plurality of array signal lines **4** are respectively arranged in each insulating layer, for example, the signal lines **4** transmit data signals, and the plurality of signal lines **4** are respectively insulated by the insulating layers. In the embodiment, the signal lines **4** may be arranged in a same layer with the gate metal layer in the display area.
(81) In addition, a power metal line **5** is further arranged above the signal line **4** arranged in the first transition sub-area **201**, and a first blocking structure **410** and a second blocking structure **420** are formed on the power metal line **5**. The first blocking structure **410** and the second blocking structure **420** respectively include a third planarization layer **411**, a third pixel defining layer **412**, and an encapsulation layer **413** covering the blocking structures, which are sequentially stacked from bottom to top. Wherein the second barrier structure **420** includes two stacked third planarization layers **411** to increase the height of the second barrier structure **420** compared to the first barrier structure **410** closer to the display area **100**.
(82) Optionally, in the embodiment of the present disclosure, the third planarization layer **411** in the first barrier structure **410** and the second barrier structure **420** is arranged in the same layer as the first planarization layer **141** or the second planarization layer **142** of the display area **100**; the third pixel defining layer **412** in the first and second barrier structures **410** and **420** is arranged in the same layer as the first pixel defining layer **7031** of the display area **100**; the encapsulation layer **413** in the first barrier structure **410** and the second barrier structure **420** is arranged at the same layer as the spacer layer **706** of the display area **100**.
(83) Further, optionally, the first transition sub-area **201** further includes a buffer layer **430** sequentially located on the first blocking structure **410** and the second blocking structure **420**, a touch wiring layer **900** located on the buffer layer **430**, and a second organic layer **501** located on the touch wiring layer **900**. Optionally, the buffer layer **430** may be an inorganic layer, or may have a structure in which an inorganic layer is stacked over an organic layer.
(84) Optionally, the touch wiring layer **900** includes a first touch trace **901**, an insulating layer **902**, and a second touch trace **903** arranged in sequence. One of the first touch trace **901** and the second touch trace **903** is connected to the first touch electrode **921** of the display area **100**, and the other of the first touch trace **901** and the second touch trace **903** is connected to the second touch electrode **922** of the display area **100**.
(85) In the embodiment of the present disclosure, optionally, one of the first touch lines **901** and

one of the second touch lines **903** form a group, and are connected in parallel or directly connected by via holes in a direction perpendicular to the substrate **1**, so as to reduce the trace resistance. The first touch lines **901** and the second touch lines **903** connected as a group are respectively connected to the touch electrodes and the sensing electrodes of the display area **100** in a one-to-one correspondence manner.

(86) Optionally, the second organic layer **501** of the first transition sub-area **201** is arranged in the same layer as the first organic layer **920** of the display region **100**.

(87) Further, in the embodiment of the present disclosure, the bending sub-area **202** includes: the liquid crystal display comprises a substrate **1**, and a fourth planarization layer **213**, a metal wiring layer **211**, a fifth planarization layer **214**, a fourth pixel definition layer **215**, a spacer layer **216** and a second organic layer **501** which are sequentially arranged on the substrate **1**.

(88) Optionally, the fourth planarization layer **213** of the bending sub-area **202** and the first planarization layer **141** of the display region **100** are arranged on the same layer, the metal wiring layer **211** of the bending sub-area **202** and the transfer metal layer **140** of the display region **100** are arranged on the same layer, and the metal wiring layer **211** is connected to the second touch circuit **902** of the first transition sub-area **201**; the fifth planarization layer **214** of the bending sub-area **202** is arranged on the same layer as the second planarization layer **142** of the display region **100**; the fourth pixel defining layer **215** of the bending sub-area **202** is arranged in the same layer as the first pixel defining layer **7031** or the second pixel defining layer **7032** of the display region **100**; the spacer layer **216** of the bending sub-area **202** is arranged on the same layer as the second pixel defining layer **7032** of the display region **100**; the second organic layer **501** of the bending sub-area **202** is an extension layer of the second organic layer **510** of the first transition sub-area **201**, and is arranged in the same layer as the first organic layer **920** of the display region **100**.

(89) In the above implementation structure of the present disclosure, the metal wiring layer **211** is a transfer metal layer for inputting a touch signal to the second touch line **902**. In addition, since the second organic layer **501** is arranged on the bending sub-area **202** away from the substrate **1**, the second organic layer **501** is formed as a thickness adjusting layer of the bending sub-area **202**, and as shown in FIG. **3**, the arrangement of the second organic layer **501** can make the vertical distance h from the central plane **2111** of the metal wiring layer **211** to the bending neutral layer **212** smaller than a preset distance, so as to ensure that the metal wiring layer **211** is arranged as close to the bending neutral layer **212** as possible, thereby reducing the stress of the metal wiring layer **211** on the bending sub-area **202** and avoiding a fracture of the metal wiring layer **211**.

(90) Further, with reference to FIG. **13**, in the embodiment of the present disclosure, the binding sub-area **204** includes a substrate **1**, a blocking layer **2**, a buffer layer **3**, at least two insulating layers, a power metal line **5**, a metal wiring layer **211**, a second planarization layer **142**, a buffer layer **225**, a first touch line **901**, an isolation layer **902**, and a second touch line **902**, which are respectively extended from the corresponding layers of the display region **100**, the first transition sub-area **201**, and the bending sub-area **202**.

(91) Based on the above embodiment, on the substrate, from the peripheral region close to the display region to the bending sub-area, the touch circuit is wired in double layers, the layers of the bending sub-area are replaced by the metal wiring layer, and after passing through the bending sub-area, the double layers of touch circuit are continuously wired in the binding sub-area. In addition to the way of disposing the second organic layer **501** on the bending sub-area **202** to form the thickness adjusting layer of the bending sub-area **202**, so that the metal wiring layer **211** is arranged as close to the bending neutral layer **212** as possible, the flexible display panel according to the embodiments of the present disclosure may further include a substrate **1** as the thickness adjusting layer, so that the substrate **1** includes at least four layers of layer structures, as shown in FIG. **11**, the substrate **1** may include a first PI material **11**, a second PI material **12** and at least two inorganic layers **13** between the first PI material and the second PI material, which are arranged oppositely, and by changing the structural composition and thickness of the substrate **1**, the overall thickness of

the bending sub-area **202** may be adjusted, so as to achieve the purpose of adjusting the position of the metal wiring layer **211** relative to the bending neutral layer **212**.

(92) In one embodiment of the present disclosure, as shown in FIG. **13**, the thickness adjusting layer (the second organic layer **501**) on the bending sub-area **202** may extend to the display region **100** and cover a side of the touch wiring layer away from the substrate.

(93) In the flexible display panel according to the above embodiment of the present disclosure, the second organic layer **501** is arranged on the bending sub-area **202** for adjusting the overall thickness of the bending sub-area **202**, where in an embodiment, as shown in FIG. **14**, the second organic layer **501** may extend to the second transition sub-area **203** and the binding sub-area **204** in addition to the first transition sub-area **201**, and covers the second transition sub-area **203** and the binding sub-area **204** on the surface away from the substrate **1**.

(94) In another embodiment of the flexible display panel of the present disclosure, optionally, according to the description of the above embodiments of the flexible display panel, the bending sub-area **202** is not limited to be able to adjust the position of the metal wiring layer **211** relative to the bending neutral layer **212** only by disposing a thickness adjusting layer on the outermost surface far away from the substrate **1**, and as shown in FIG. **8**, the bending sub-area **202** may also adjust the position of the metal wiring layer **211** relative to the bending neutral layer **212** by disposing a thickness adjusting layer between the first organic layer and the metal wiring layer, or between the second organic layer and the metal wiring layer.

(95) Based on this embodiment, when the surface of the bending sub-area **202** away from the substrate **1** does not need to be with the thickness adjusting layer, that is, the second organic layer **501** is not provided, the structure of the flexible display panel in the non-display region is as shown in FIG. **15**.

(96) It should be noted that, in the present disclosure, the different film layers mentioned are the same layer, that is, the two film layers are made by the same composition process, or are prepared on the same film layer.

(97) An embodiment of the present disclosure further provides a display device, where the display device includes the flexible display panel described in any one of the above.

(98) In the embodiment of the disclosure, on the non-display area, a driving component is arranged at a side of the bending sub-area, which is far away from the display area, and the metal wiring layer is electrically connected to the driving component; the bending sub-area is bent relative to the display area, and the driving component is arranged at a side away from the display area.

(99) Optionally, the driving component includes a driving chip and a flexible printed circuit board, and as shown in FIG. **2**, by bending the bending sub-area **202** toward a side away from the display region **100** with respect to the display region **100**, the driving component **300** located on the non-display region **200** and far away from the display region **100** may be bent to the back of the display region **100**, so as to complete the next step of module assembly, thereby achieving the effects of reducing the display frame of the display region **100** of the flexible display panel and realizing narrow-frame display.

(100) According to the flexible display panel and the display device, the vertical distance h from the metal wiring layer to the bending neutral layer is smaller than the preset distance, so that the metal wiring layer is arranged close to the bending neutral layer as much as possible, the stress of the metal wiring layer on the bending sub-area is reduced, the metal wiring layer is prevented from being broken, and the yield of the bending sub-area is improved.

(101) Furthermore, the thickness adjusting layer is arranged at the position, far away from the substrate, of the bending sub-area, so that the vertical distance h from the metal wiring layer to the bending neutral layer is smaller than the preset distance.

(102) Another aspect of the present disclosure also provides a method for forming the flexible display panel, where the method includes: providing a substrate; forming a display area and a non-display area on the substrate; the non-display area comprises a bending sub-area and a binding sub-

area, and the bending sub-area is configured to bend the binding sub-area to a side away from the display area; the display area comprises a driving circuit layer, the driving circuit layer comprises a source/drain electrode layer, a planarization layer arranged at a side, far away from the substrate, of the source/drain electrode layer, a pixel defining layer arranged at a side, far away from the substrate, of the planarization layer and a touch wiring layer arranged at a side, far away from the substrate, of the pixel defining layer, the bending sub-area comprises a first organic layer, a second organic layer and a metal wiring layer, the first organic layer and the second organic layer are arranged on the substrate, the metal wiring layer is located between the first organic layer and the second organic layer, the touch wiring layer is electrically connected to the metal wiring layer, and the orthographic projection of the touch wiring layer on the plane of the substrate is not located in the bending sub-area; when the bending sub-area is in a bending state, the vertical distance from the bending neutral layer of the bending sub-area to the metal wiring layer is smaller than a preset distance.
(103) Optionally, the metal wiring layer and the source/drain electrode layer are arranged in the same layer when the display region and the non-display region are formed.
(104) Optionally, when the display region and the non-display region are formed, the display region further comprises a first electrode arranged in the pixel defining layer, a light emitting layer arranged at a side of the first electrode away from the substrate, and a second electrode arranged at a side of the light emitting layer away from the substrate; the first electrode is connected to the source/drain electrode layer through a transfer metal layer, and the metal wiring layer and the transfer metal layer are arranged on the same layer.
(105) While the foregoing is directed to embodiments of the present disclosure, it will be appreciated by those skilled in the art that various changes and modifications may be made without away from the principles of the disclosure, and it is intended that such changes and modifications be considered as within the scope of the disclosure.

Claims

1. A flexible display panel, comprising a display area and a non-display area, wherein the non-display area comprises a bending sub-area and a binding sub-area, and the bending sub-area is configured to bend the binding sub-area to a side away from the display area; the display area comprises a driving circuit layer arranged on a substrate, the driving circuit layer comprises a source/drain electrode layer, a planarization layer arranged at a side, far away from the substrate, of the source/drain electrode layer, a pixel defining layer arranged at a side, far away from the substrate, of the planarization layer and a touch wiring layer arranged at a side, far away from the substrate, of the pixel defining layer, the bending sub-area comprises a first organic layer, a second organic layer and a metal wiring layer between the first organic layer and the second organic layer arranged on the substrate, the touch wiring layer is electrically connected to the metal wiring layer, and an orthographic projection of the touch wiring layer on a plane of the substrate is not located in the bending sub-area; wherein the second organic layer extends to the binding sub-area from the display area; wherein the binding sub-area comprises the substrate, a barrier layer, a first buffer layer, at least two insulating layers, a power metal line, the metal wiring layer, a second planarization layer, a second buffer layer, a first touch trace, an isolation layer and a second touch trace which are respectively extended from corresponding layers of the display region, the first transition sub-area and the bending sub-area.
2. The flexible display panel according to claim 1, wherein an orthographic projection of the second organic layer overlaps with an orthographic projection of the binding sub-area.
3. The flexible display panel according to claim 1, wherein the metal wiring layer is arranged in a same layer as the source/drain electrode layer.
4. The flexible display panel according to claim 1, wherein the display region further comprises a

first electrode within the pixel defining layer, a light-emitting layer at a side of the first electrode away from the substrate and a second electrode at a side of the light-emitting layer away from the substrate; the first electrode is connected to the source/drain electrode layer through a transfer metal layer, and the metal wiring layer and the transfer metal layer are arranged on the same layer.
5. The flexible display panel according to claim 4, wherein the planarization layer comprises a first planarization layer and the second planarization layer, wherein the transfer metal layer is on the first planarization layer and within the second planarization layer, and the transfer metal layer is connected to the source/drain electrode layer through a first via hole penetrating through the first planarization layer; the first electrode is connected to the transfer metal layer through a second via hole penetrating through the second planarization layer.
6. The flexible display panel according to claim 5, wherein the first organic layer is closer to the substrate than the second organic layer, the first organic layer is in the same layer as the first planarization layer, the second organic layer is in the same layer as the second planarization layer and the pixel defining layer.
7. The flexible display panel according to claim 1, in response to the bending sub-area being in a bending state, a vertical distance from a bending neutral layer of the bending sub-area to the metal wiring layer is smaller than a preset distance.
8. The flexible display panel according to claim 7, wherein the bending neutral layer overlaps the metal wiring layer in response to the bending sub-area being in a bent state.
9. The flexible display panel according to claim 1, wherein the metal wiring layer is located at one-half of a thickness of the bending sub-area in response to the bending sub-area being not in the bent state.
10. The flexible display panel according to claim 7, wherein a thickness adjusting layer is at a side of the second organic layer away from the metal wiring layer in the bending sub-area; through the thickness adjusting layer, in response to the bending sub-area being in a bending state, a vertical distance from the bending neutral layer of the bending sub-area to the metal wiring layer is smaller than a preset distance.
11. The flexible display panel according to claim 10, wherein the thickness adjusting layer further extends to the display area and is located at a side of the touch wiring layer away from the substrate.
12. The flexible display panel according to claim 10, wherein the thickness adjusting layer is an organic material layer.
13. The flexible display panel according to claim 1, wherein the substrate comprises a first organic material, a second organic material, and at least one inorganic layer between the first organic material and the second organic material.
14. The flexible display panel according to claim 7, wherein the bending neutral layer is located at a side of the metal wiring layer proximate to the substrate.
15. The flexible display panel according to claim 1, wherein a preset distance is less than or equal to 5 microns.
16. A display device comprising a flexible display panel, wherein the flexible display panel comprises: a display area and a non-display area, wherein the non-display area comprises a bending sub-area and a binding sub-area, and the bending sub-area is configured to bend the binding sub-area to a side away from the display area; the display area comprises a driving circuit layer arranged on a substrate, the driving circuit layer comprises a source/drain electrode layer, a planarization layer arranged at a side, far away from the substrate, of the source/drain electrode layer, a pixel defining layer arranged at a side, far away from the substrate, of the planarization layer and a touch wiring layer arranged at a side, far away from the substrate, of the pixel defining layer, the bending sub-area comprises a first organic layer, a second organic layer and a metal wiring layer between the first organic layer and the second organic layer arranged on the substrate, the touch wiring layer is electrically connected to the metal wiring layer, and an orthographic

projection of the touch wiring layer on a plane of the substrate is not located in the bending sub-area; wherein the second organic layer extends to the binding sub-area from the display area; wherein the binding sub-area comprises the substrate, a barrier layer, a first buffer layer, at least two insulating layers, a power metal line, the metal wiring layer, a second planarization layer, a second buffer layer, a first touch trace, an isolation layer and a second touch trace which are respectively extended from corresponding layers of the display region, the first transition sub-area and the bending sub-area.

17. The display device according to claim 16, wherein a driving component is arranged on the binding sub-area on the non-display region, and the metal wiring layer is electrically connected to the driving component; the bending sub-area is bent relative to the display area, and the driving component is arranged at a side away from the display area.

18. A method of forming a flexible display panel, comprising: providing a substrate; and forming a display area and a non-display area on the substrate; wherein the non-display area comprises a bending sub-area and a binding sub-area, and the bending sub-area is configured to bend the binding sub-area to a side away from the display area; the display area comprises a driving circuit layer arranged on a substrate, the driving circuit layer comprises a source/drain electrode layer, a planarization layer arranged at a side, far away from the substrate, of the source/drain electrode layer, a pixel defining layer arranged at a side, far away from the substrate, of the planarization layer and a touch wiring layer arranged at a side, far away from the substrate, of the pixel defining layer, the bending sub-area comprises a first organic layer, a second organic layer and a metal wiring layer between the first organic layer and the second organic layer arranged on the substrate, the touch wiring layer is electrically connected to the metal wiring layer, and an orthographic projection of the touch wiring layer on a plane of the substrate is not located in the bending sub-area; wherein the second organic layer extends to the binding sub-area from the display area; wherein the binding sub-area comprises the substrate, a barrier layer, a first buffer layer, at least two insulating layers, a power metal line, the metal wiring layer, a second planarization layer, a second buffer layer, a first touch trace, an isolation layer and a second touch trace which are respectively extended from corresponding layers of the display region, the first transition sub-area and the bending sub-area.